Exercise 3.15

Multiple choice questions

1. If $x^3 + 6x^2 + kx + 6$ is exactly divisible by $(x + 2)$, then k= ?
   (1) –6 (2) –7 (3) –8 (4) 11

2. The root of the polynomial equation $2x + 3 = 0$ is
   (1) $\frac{1}{3}$ (2) $-\frac{1}{3}$ (3) $-\frac{3}{2}$ (4) $-\frac{2}{3}$

3. The type of the polynomial $4–3x^3$ is
   (1) constant polynomial (2) linear polynomial
   (3) quadratic polynomial (4) cubic polynomial.

4. If $x^{51} + 51$ is divided by $x + 1$, then the remainder is
   (1) 0 (2) 1 (3) 49 (4) 50

5. The zero of the polynomial $2x+5$ is
   (1) $\frac{5}{2}$ (2) $-\frac{5}{2}$ (3) $\frac{2}{5}$ (4) $-\frac{2}{5}$

6. The sum of the polynomials $p(x) = x^3 – x^2 – 2$, $q(x) = x^2–3x+ 1$
   (1) $x^3 – 3x – 1$ (2) $x^3 + 2x^2 – 1$ (3) $x^3 – 2x^2 – 3x$ (4) $x^3 – 2x^2 + 3x –1$

7. Degree of the polynomial $(y^3–2)(y^3 + 1)$ is
   (1) 9 (2) 2 (3) 3 (4) 6

8. Let the polynomials be
   (A) $–13q^5 + 4q^2 + 12q$ (B) $(x^2 +4)(x^2 + 9)$
   (C) $4q^8 – q^6 + q^2$ (D) $-\frac{5}{7}y^{12} + y^3 + y^5$
   Then ascending order of their degree is
   (1) A,B,D,C (2) A,B,C,D (3) B,C,D,A (4) B,A,C,D

9. If $p(a) = 0$ then $(x - a)$ is a _____ of $p(x)$
   (1) divisor (2) quotient (3) remainder (4) factor

10. Zeros of $(2 - 3x)$ is _____
    (1) 3 (2) 2 (3) $\frac{2}{3}$ (4) $\frac{3}{2}$

11. Which of the following has $x - 1$ as a factor?
    (1) $2x - 1$ (2) $3x - 3$ (3) $4x - 3$ (4) $3x - 4$

12. If $x - 3$ is a factor of $p(x)$, then the remainder is
    (1) 3 (2) –3 (3) $p(3)$ (4) $p(–3)$

13. $(x + y)(x^2 - xy + y^2)$ is equal to
    (1) $(x + y)^3$ (2) $(x - y)^3$ (3) $x^3 + y^3$ (4) $x^3 - y^3$

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